

Week 4, Lecture 1: Combinatorics, Distributions, and Expectation

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker 4.4 (analyzing probability distributions, p. 155-159). Combinatorics/binomial-coefficient derivation is supplementary (not in Bolker’s own text, which just states the formula) – built from the WOD’s lottery problem and the idea bank’s Casella & Berger reference. Also mined: the Week 3 Lecture 2 notes (density-vs-probability framing, $\text{Var}(X)=E[X^2]-\mu^2$ derivation) and the `WOD_Week3_Lecture2_ProbabilityPuzzle_Key.Rmd` birthday-problem derivation.

Where this sits in the week

Today’s WOD reuses last week’s probability puzzle (Week 3, Lecture 2 – students didn’t get to it live, since Wednesday’s lecture ran out of time before the birthday-problem teaser could be revealed). That reuse isn’t incidental: Part 2 of that WOD (the lottery-ticket problem) and Part 3 (the birthday teaser) are both combinatorics problems, and today’s lecture opens by formalizing the counting tools that problem set only asked students to feel their way through.

- Last Wednesday (Week 3, Lecture 2): axioms of probability, joint/ marginal/conditional probability, Bayes’ rule.
- **Today:** formalize counting (permutations, combinations, the binomial coefficient), resolve the birthday-problem teaser as the worked example, then build up random variables, probability distributions (pmf vs. pdf), and expectation/variance from first principles.
- **This Wednesday (Week 4, Lecture 2):** the bestiary of distributions, with today’s expectation/variance machinery reused immediately in the WOD deriving $\text{Var}(\bar{X})$. Synthetic data simulation (where these distributions become the stochastic half of a full data-generating model) moved to next Monday (Week 5, Lecture 1).

Rationale for the flow

This is a dense lecture – four sub-topics in one 30-35 minute period – so keep each one tight and lean on the WOD callback to avoid re-deriving things from zero. Students already computed a binomial pmf by hand in Lab 3 (Act 2, using $\binom{N}{k}$ as a given formula) and already have Bayes’ rule and the axioms from last week; today’s job is to open the black box on $\binom{N}{k}$ itself, then build the vocabulary (random variable, distribution, pmf/pdf, expectation, variance) that the rest of the semester leans on constantly.

The birthday problem is doing double duty: it’s the payoff students were promised last week (don’t skip it – it’s the single best “wait, really?” moment available this week), and it’s also the natural worked example for combinatorics, since the reveal explicitly needs $\binom{25}{2}$ to explain *why* the answer is so counterintuitive.

Part A: Counting – permutations, combinations, and $\binom{N}{k}$ (~10-12 min)

Motivating question, recap from the WOD: the lottery problem asked how many 5-number tickets are possible from 1-30, when order doesn’t matter. Ask students what they got before revealing the general recipe.

Permutations first (order matters). How many ways can you pick and *arrange* 5 distinct numbers out of 30? First pick: 30 choices. Second pick: 29 remain. Third: 28. Etc.

$$30 \times 29 \times 28 \times 27 \times 26 = \frac{30!}{25!}$$

In general, the number of ordered selections of k items from N is

$$P(N, k) = \frac{N!}{(N - k)!}.$$

Combinations (order doesn’t matter). A lottery ticket $\{3, 8, 14, 20, 29\}$ is the same ticket no matter what order you wrote the numbers down in. Every unordered ticket got counted once for each of the $5! = 120$ orders its 5 numbers could have been written in – so the permutation count *overcounts* by a factor of $k!$. Divide it out:

$$\binom{N}{k} = \frac{P(N, k)}{k!} = \frac{N!}{k! (N - k)!}$$

```
choose(30, 5)
```

```
## [1] 142506
```

Sticking point to flag: this is the single most common combinatorics error – forgetting to divide by $k!$, i.e. answering with the permutation count instead of the combination count.

Worth asking directly: “why divide by 5! and not, say, 4! or 6!?” (Answer: because there are exactly 5! orderings of a 5-element set.)

Tie back to the binomial pmf, which Lab 3 already used as a given formula:

$$P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$$

Now every piece of this formula has been derived from scratch: $\binom{N}{k}$ counts *how many* of the N trials could be the k successes, and $p^k (1 - p)^{N-k}$ is the probability of any *one* specific arrangement of k successes and $N - k$ failures.

Part B: The birthday problem, revealed (~8 min)

Callback: “last week I asked you to guess whether 25 people in a room have better than 50-50 odds of sharing a birthday. Who guessed yes? No?” Most students guessed no. Walk through the full derivation (complete version in WOD_Week3_Lecture2_ProbabilityPuzzle_Key.Rmd):

$$P(\text{at least one match}) = 1 - P(\text{no matches at all})$$

Building “no matches” one person at a time (each new person must avoid all previously-assigned birthdays):

$$P(\text{no matches}) = \prod_{k=1}^{25} \frac{365 - k}{365}$$

```
n <- 25
p_no_match <- prod((365 - 0:(n - 1)) / 365)
p_match <- 1 - p_no_match
p_match
```

```
## [1] 0.5686997
```

56.9% – comfortably past the 50% line.

Why this feels wrong, and why it’s today’s example (not just a fun fact): the intuitive comparison is “25 people vs. 365 days,” but the real driver is the number of *pairs* of people, since any pair could be the one that matches:

```
choose(25, 2)
```

```
## [1] 300
```

25 people make $\binom{25}{2} = 300$ distinct pairs – that’s a direct application of the combination formula from Part A, and it’s *why* 300 independent-ish chances at a 1-in-365 coincidence compounds into better than even odds. This is the payoff for teaching combinatorics before probability distributions: the binomial coefficient isn’t just an input to a pmf formula, it’s a general counting tool that shows up anywhere you need to count possibilities.

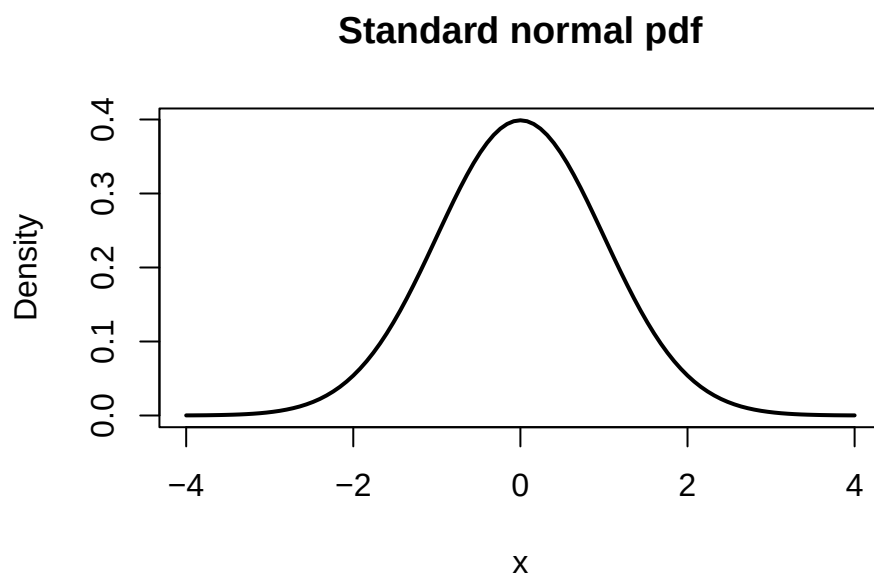
Part C: Random variables and probability distributions (~8 min)

A **random variable** X maps outcomes onto numbers. A **probability distribution** assigns a probability (discrete) or density (continuous) to every value in X 's support.

Discrete: probability mass function (pmf). $P(X = x)$ for each possible value x . Must be non-negative and sum to 1 over the support. The binomial pmf from Part A is one example; write it on the board again, now fully derived.

Continuous: probability density function (pdf). $f(x)$, a curve rather than a bar chart. Must integrate to 1 over the support.

```
curve(dnorm(x, mean = 0, sd = 1), from = -4, to = 4, lwd = 2,  
      xlab = "x", ylab = "Density", main = "Standard normal pdf")
```



Sticking point to flag – density $\neq$ probability: the height of $f(x)$ is *not* itself a probability – it can exceed 1 (try `dnorm(0, 0, 0.1)`, a normal with a very small standard deviation). Only the **area under the curve** between two values is a probability:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Whereas for a discrete pmf, $P(X = x)$ *is* directly a probability – you can read it straight off the bar chart. This distinction (mass you can read directly vs. density you have to integrate) is the core difference between the two kinds of distributions.

```
dnorm(0, mean = 0, sd = 0.1)  # > 1 -- a density, not a probability
```

```
## [1] 3.989423
```

Part D: Expectation and variance (~10 min)

Expectation $E[X]$ is the long-run average value of X – a weighted average of every possible outcome, weighted by its probability (or density):

$$E[X] = \sum_x x P(X = x) \quad (\text{discrete}) \quad E[X] = \int x f(x) dx \quad (\text{continuous})$$

Physical intuition: $E[X]$ is the “center of mass” of the distribution – if you cut the pmf/pdf out of cardboard, $E[X]$ is where it balances on a fingertip.

Linearity of expectation (true always, no independence required):

$$E[aX + b] = aE[X] + b \quad E[X + Y] = E[X] + E[Y]$$

This will get used immediately in Wednesday’s WOD – flag that it’s coming back.

Variance measures spread around the mean: $\text{Var}(X) = E[(X - \mu)^2]$, where $\mu = E[X]$. Worth deriving the computational formula on the board – students will use it constantly but rarely see it derived (Calc-II algebra only, no new calculus):

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

Since $E[X] = \mu$:

$$= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$$

This is just expanding a square and using linearity of expectation – nothing harder than what Week 1’s algorithms lecture already assumed.

Variance scaling: $\text{Var}(aX) = a^2 \text{Var}(X)$ – note the squared coefficient (variance doesn’t scale linearly, since it’s built from squared deviations). Also flag this as a preview for Wednesday’s WOD.

Take-home message

- $\binom{N}{k}$ counts unordered subsets; divide the permutation count by $k!$ to remove the overcounting from order.
- The birthday problem is counterintuitive because the real driver is the number of *pairs* ($\binom{25}{2} = 300$), not the number of people.
- A pmf gives probabilities directly; a pdf gives densities, and only an *area* under a pdf is a probability.
- $E[X]$ is a weighted average (linear); $\text{Var}(X) = E[X^2] - \mu^2$ measures spread and scales as a^2 , not a .

Roadmap tie-in

- This Wednesday's WOD picks up immediately where Part D leaves off, using linearity of expectation and variance scaling to derive $\text{Var}(\bar{X}) = \text{Var}(X)/n$.
- Next Monday's lecture (synthetic data simulation) uses these distributions as the stochastic half of a full data-generating model.

Discussion seed (if time remains)

Ask: *Lab 3's binomial and Poisson pmfs, and the normal pdf coming up in next week's synthetic-data lecture – can you think of an ecological quantity that would need a pmf (discrete) vs. one that would need a pdf (continuous)? What about a quantity that's naturally continuous but that ecologists often round to a discrete count anyway (e.g., body counts from a density estimate)?*

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

0. Open (~2 min)

- Callback: “who guessed yes/no on the birthday problem last week?” – today gives you the tools to answer it for real

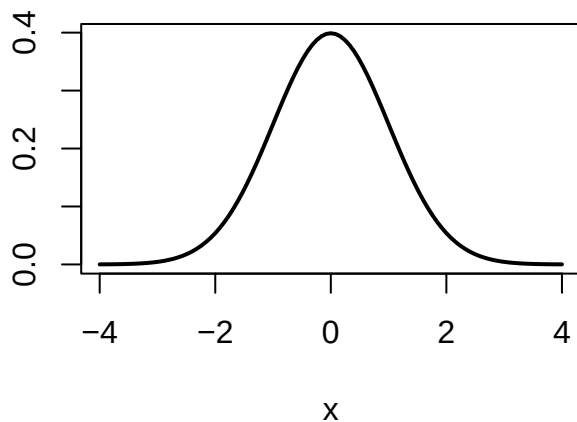
1. Counting: permutations $\rightarrow$ combinations (~10-12 min)

- Permutations (order matters): $P(N, k) = \frac{N!}{(N - k)!}$
- Combinations (order doesn't): divide out the $k!$ overcounting – $\binom{N}{k} = \frac{N!}{k!(N - k)!}$
- **Sticking point:** forgetting to divide by $k!$
- Ties to binomial pmf: $\binom{N}{k}$ counts *which* trials succeed; $p^k(1 - p)^{N-k}$ is the probability of *one* such arrangement

2. Birthday problem, revealed (~8 min)

- $P(\text{match}) = 1 - \prod_{k=1}^{25} \frac{366-k}{365} \approx 0.569$ – past 50-50!
- **Why:** the real driver is $\binom{25}{2} = 300$ pairs, not 25 people vs. 365 days

3. Random variables and distributions: pmf vs. pdf (~8 min)



- Discrete pmf: $P(X = x)$ read directly off the bar chart, sums to 1

- Continuous pdf: $f(x)$, a curve; only *area* is a probability
- **Sticking point:** density $\neq$ probability – $f(x)$ can exceed 1

4. Expectation and variance (~10 min)

- $E[X] = \sum xP(X = x)$ or $\int xf(x)dx$ – weighted average / center of mass
- Linearity (always true): $E[aX + b] = aE[X] + b$, $E[X + Y] = E[X] + E[Y]$
- **Derive on board:** $\text{Var}(X) = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$
- Scaling: $\text{Var}(aX) = a^2\text{Var}(X)$ – squared, not linear

Discussion (if time)

- *Which ecological quantities need a pmf? Which need a pdf? Any that are naturally continuous but usually treated as discrete?*